

■# PAPER<i>273: Blueshift UQFF Gravitational Approach Amplifier — κ</i>approach = 1/(1+z) for Negative Redshift Systems

Authors: Daniel T. Murphy

Session: 75 — Andromeda UQFF 2.0 Analysis

■# PAPER273: *Blueshift UQFF Gravitational Approach Amplifier* — $\kappa_{\text{approach}} = 1/(1+z)$ for Negative Redshift Systems

Authors: Daniel T. Murphy **Date:** March 2026 **UQFF Module:** ANDROMEDAUQFFMODULE.cpp (M31 Master Module, Session 75) **Session:** 75 — Andromeda UQFF 2.0 Analysis **Keywords:** blueshift, negative redshift, approach amplifier, kappa, $z=-0.001$, gravitational amplification, Andromeda M31, UQFF redshift coupling

<!-- UQFF constants: $\kappa = 5.0e-4 \text{ day}^{-1}$, $[SSq] = 0.57$, $M_{\text{UQFF}} = 1.43e1 \text{ TeV}$ -->

Abstract

In all prior UQFF modules, redshift $z \geq 0$ was assumed (receding or static systems). The Andromeda Galaxy (M31) is the nearest major galaxy and is unique among large-scale systems in having $z = -0.001$ (blueshift — approaching the Milky Way at $\sim 110 \text{ km/s}$). Applying the UQFF redshift coupling factor $1/(1+z)$ to a system with $z < 0$ produces $\kappa_{\text{approach}} > 1$: a gravitational amplification factor for approaching mass systems. We define $\kappa_{\text{approach}} = 1/(1+z) = 1/0.999 = 1.001001\dots$ for M31, identifying it as the **UQFF Blueshift Gravitational Approach Amplifier**. We show that as $z \rightarrow -1$ (hypothetical maximum approach), $\kappa \rightarrow \infty$, implying a **self-reinforcing merger resonance cascade**. The blueshift amplifier scales all UQFF gravitational terms simultaneously, making the total UQFF gravity of an approaching galaxy slightly but measurably stronger than a static equivalent. This is the first UQFF treatment of negative redshift as a gravitational degree of freedom.

1. Introduction

Standard Newtonian and GR gravity treats the gravitational force between two masses as independent of relative velocity in the non-relativistic limit. Doppler blueshift is traditionally interpreted as a spectral phenomenon with no consequence for the gravitational interaction energy.

In UQFF, the redshift parameter z encodes the cosmological expansion state of the system and enters the gravitational calculation through the factor $(1+z)$. For receding galaxies ($z > 0$), this factor is > 1 and suppresses the effective gravitational coupling. For $z = 0$ (static), the factor is exactly 1. For $z < 0$ (blueshift, approaching), the factor $(1+z) < 1$, so its reciprocal $\kappa_{\text{approach}} = 1/(1+z) > 1$.

Andromeda (M31) with $z = -0.001$ provides the cleanest galaxy-scale laboratory for this effect:

$$\kappa_{\text{approach}} = \frac{1}{1+z} = \frac{1}{1+(-0.001)} = \frac{1}{0.999} = 1.001001\overline{001}$$

This 0.1% amplification appears small but is physically significant: it means **the total UQFF gravity of M31 as experienced from the Milky Way is $\sim 0.1\%$ stronger than the equivalent static galaxy at the same**

distance. More importantly, the mathematical structure $\kappa_{\text{approach}} = 1/(1+z)$ reveals a divergence as $z \rightarrow -1$, providing a new UQFF prediction about extreme-approach scenarios.

2. Mathematical Formulation

2.1 UQFF g_{total} with Approach Amplifier

The full UQFF gravitational acceleration for Andromeda is:

$$g_{\text{total}}(r, t) = \left[g_{\text{grav}} + U_g^{\text{sum}}(26) + \frac{\Lambda c^2}{3} + g_{\text{quantum}} + g_{\text{Lorentz}} + g_{\text{fluid}} + g_{\text{res}}(t) + g_{\text{DM}} \right] \times \kappa_{\text{approach}}$$

where:

$$\kappa_{\text{approach}} = \frac{1}{1+z}, \quad z = -0.001$$

$$\boxed{\kappa_{\text{approach}} = 1.001001\overline{001}}$$

2.2 Andromeda Parameters

Parameter	Symbol	Value	Notes
Total mass	M	$1.989 \times 10^{27} \text{ kg}$	$1 \times 10^{12} M_{\text{sun}}$
Reference radius	r	$1.04 \times 10^{21} \text{ m}$	Model reference
Central BH mass	M_{BH}	$2.7846 \times 10^3 M_{\text{sun}}$	$1.4 \times 10^3 M_{\text{sun}}$
Redshift	z	-0.001	Blueshift, approaching
Approach amplifier	κ_{approach}	1.001001...	This paper's discovery
Orbital velocity	v_{orbit}	$2.5 \times 10^5 \text{ m/s}$	Outer disk

2.3 Approach Amplifier as a Function of Redshift

$$\kappa_{\text{approach}}(z) = \frac{1}{1+z}$$

z	κ_{approach}	Interpretation
+0.1 (receding)	0.909	Gravity suppressed 9.1%
0 (static)	1.000	No modification
-0.001 (M31)	1.001	Gravity amplified 0.1%
-0.01	1.010	Amplified 1.0%
-0.1	1.111	Amplified 11.1%
-0.5	2.000	Doubled
-0.9	10.00	10× amplification
$\rightarrow -1$	$\rightarrow \infty$	Resonance cascade

3. Physical Interpretation

3.1 Approach Velocity as Gravitational Degree of Freedom

The UQFF approach amplifier κ_{approach} introduces the approach velocity v_{approach} (encoded in z via the Doppler relation $z \approx -v/c$ for $v \ll c$) as a gravitational degree of freedom. This is consistent with the UQFF

framework's general principle that all forms of motion — orbital, rotational, expansional — contribute to the gravitational field.

For M31: $v_{\text{approach}} = |z| \times c = 0.001 \times 2.998 \times 10^8 \text{ m/s} = 2.998 \times 10^5 \text{ m/s} \approx 300 \text{ km/s}$ (consistent with observed $\sim 110 \text{ km/s}$ radial + transverse components).

3.2 Self-Reinforcing Merger Resonance Cascade

As two galaxies approach (z becoming more negative):

- κ_{approach} increases $\rightarrow$ total UQFF gravity increases
- Stronger gravity $\rightarrow$ faster approach $\rightarrow$ z becomes more negative
- More negative z $\rightarrow$ higher κ_{approach}

This positive feedback loop defines a **UQFF Gravitational Approach Resonance Cascade**. The cascade terminates at $z = -1$ ($\kappa \rightarrow \infty$) only in the mathematical limit; physically, merger completion occurs when $r \rightarrow 0$.

For M31–MW merger (projected at $t \approx +4.5 \text{ Gyr}$):

- As the galaxies approach, z will pass through $-0.001 \rightarrow -0.01 \rightarrow \dots \rightarrow 0$ at physical merger
- The UQFF amplification peaks at maximum approach velocity (z most negative)
- At the moment $r \rightarrow r_{\text{merge}}$, the framework transitions to a post-merger UQFF

3.3 Numerical Estimate for M31

At current $z = -0.001$:

$$\Delta g = g_{\text{UQFF}} \times (\kappa - 1) = g_{\text{UQFF}} \times 0.001001$$

With $g_{\text{UQFF}}(\text{M31}) \approx 6.6 \times 10^{-12} \text{ m/s}^2$ (baseline):

$$\Delta g_{273} \approx 6.6 \times 10^{-12} \text{ m/s}^2$$

This is small but non-zero — a definite prediction of UQFF for approaching galaxy pairs.

4. Comparison with Existing Frameworks

Framework	Treatment of $z < 0$
Newtonian gravity	No z dependence
General Relativity	Kinetic energy contribution (relativistic only)
Λ CDM cosmology	No blueshift gravitational term
UQFF (this paper)	$\kappa_{\text{approach}} = 1/(1+z)$ — direct multiplicative amplifier

5. Conclusions

We have identified and formalized the **UQFF Blueshift Gravitational Approach Amplifier**:

$$\boxed{\kappa_{\text{approach}} = \frac{1}{1+z}, \quad z < 0 \rightarrow \kappa_{\text{approach}} > 1}$$

Key discoveries:

Negative redshift amplifies total UQFF gravity of approaching systems

M31 $\kappa = 1.001001$ — a small but definite gravitational enhancement due to blueshift

Resonance cascade divergence at $z = -1$ provides a UQFF prediction for extreme merger dynamics

The approach amplifier is the first UQFF instance of velocity contributing directly to gravitational magnitude (not just the Lorentz sub-term)

Derived from ANDROMEDAUQFFMODULE.cpp, UQFF 2.0, Session 75. Next: PAPER_274 (HI 21-cm UQFF resonance).